Control is Nothing Without Trust a First Look into Digital Identity Wallet Trends

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Abstract. In recent years, user-centric digital identity wallets have become increasingly available, aiming to give individuals direct control over their personal data. The EU proposal in the context of the recently revised eIDAS and its EU digital identity wallet reflects the high ambitions in this field at the EU level. However, to the best of our knowledge, no research review on the core technologies used to implement various features of these wallet solutions has been proposed so far. Our work addresses this issue providing a systematic analysis of a selection of available digital identity wallets on the market in terms of the technologies used to establish trust and controlled sharing. Building on this overview and help providers (including companies and European Member States) in charge of developing secure and trustworthy wallet-based infrastructures, we propose a classification of wallet solutions along two main dimensions (namely trust establishment and controlled credential sharing) with the goal of assisting designers to make informed design decisions and guarantee security, privacy, and trustworthiness.

Keywords: Digital Identity · Identity Wallet · Trust

1 Introduction

The digital identity landscape has evolved over time, starting with the most basic isolated identity management (IdM) systems and progressing through several stages with the addition of other models such as centralized, federated, and user-centric IdM systems. The isolated model allows service providers (SPs) to serve as identity providers (IdPs) to their users, providing unique identifiers (IDs) and associated credentials such as passwords. In the centralized model, a single IdP is used by all SPs. The shift to the federated model enables a set of SPs to recognize user IDs provided by IdPs from other SPs within a federated domain, allowing users to authenticate once and access multiple SPs without the need for re-authentication [43,71]. The three models mentioned prioritize the IdP/SPs' viewpoint, giving those entities control over the system. However,

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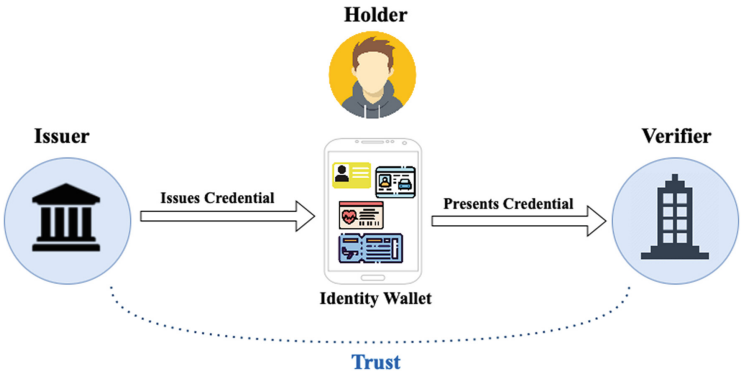

Fig. 1. Identity wallet role in a user-centric IdM system

a shift towards a user-centric model has occurred due to a growing awareness of privacy concerns and users' realization of the value of their data. This shift aims to address security and privacy issues present in the earlier models, such as data breaches resulting from IdPs aggregating large volumes of personal data [7]. User-centric IdM systems empower users with full control over their credentials, allowing them to store their credentials in storage they own or control, such as a digital identity wallet, and share them as desired [54].

A digital identity wallet is an application under the control of a user that allows the storage of multiple digital identities and related credentials. Users can use these identities to authenticate themselves online (and offline) with different levels of assurance (LoAs[1]). These identities can be obtained from government or private companies, as well as sovereign citizen documents like national identity cards and passports [47]. Figure 1 illustrates a user-centric IdM system, with a digital identity wallet at its core. The system involves three key roles: Issuer, Holder, and Verifier. The Issuer creates and issues credentials to the Holder, who receives and stores the credentials, and when required, shares them with the Verifier. The Verifier receives and verifies the credentials presented by the Holder. This system removes the need for a third-party IdP and supports a direct relationship between the Holder and an SP/Verifier, giving users full control over their data through the use of an identity wallet.

We can distinguish user-centric IdM systems based on the adopted trust model into two groups: DPKI-based and PKI-based. The former applies a decentralized public key infrastructure (DPKI) and may utilize a distributed ledger or a blockchain to create and manage decentralized identifiers (DIDs) in a decentralized manner. On the other hand, the latter employs a centralized PKI, relying on a hierarchy of trusted Certification Authorities (CAs). Our work considers both systems from a technical perspective and aims to fill the knowledge gap for designers and developers. We provide insights into the best approaches for

[1] A level of assurance refers to the certainty of a claimed identity being authentic.

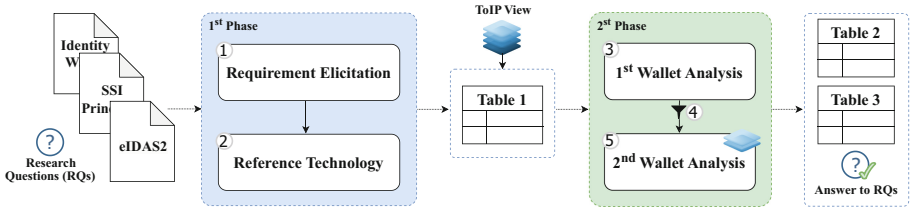

Fig. 2. Research Methodology

developing secure, private, and interoperable solutions for different use cases, based on the adopted trust model.

Organization of the Paper. First, Sect. 2 describes the research methodology employed in this study and presents the research questions. Next, Sect. 3 outlines the main requirements of digital identity wallets and the enabling technologies. This is followed by Sect. 4, which presents an analysis of a selected set of wallets in response to the research questions. Finally, Sect. 5 concludes the paper.

2 Research Methodology

The methodology we employed in this study is illustrated in Fig. 2. First of all, we pose two research questions in line with the purpose of the study which is to compare several digital identity wallet solutions made for user-centric IdM systems, based on their root of trust and the control they offer accordingly.

In the first phase: (1) we conduct a thorough literature review on digital identity wallets, eIDAS2, and SSI [35,57,60,63] to identify and extract the design requirements, both functional and non-functional, that these wallets are developed for, (2) we use next a reference technology architecture (specifically, ToIP [3]) to study the technological possibilities for these wallet solutions. As a result, we provide Table 1 with a list of non-functional requirements and the technologies that may be employed to meet them.

In the second phase, we proceed with the analysis of wallet solutions: (3) we conduct an internet search for digital identity wallets and define a set of reference wallets on the market, consisting of the first 32 wallets that provide sufficient documentation to ensure a comprehensive yet manageable analysis of the diverse landscape. We first analyze them for 6 basic differentiating features, including type (web vs mobile), use case, developer, code availability, maturity level, and supported trust model. The results of this first analysis are presented in Table 2. Then, in (4), from the reference wallets, we further narrow down our focus to a subset of 12 wallets, carefully selected to represent a fair ratio of solutions for the implemented trust models from various developers. Finally, in (5), we refer back to the ToIP and select only the technologies that address our research questions and review and study their technical specification accordingly. The results of this step are presented in Table 3 and further discussed as the answers to the research questions in Sect. 4.2.

2.1 Research Questions

We organize our analysis around two primary research questions outlined below:

RQ1. How is Trust Established? Is the Root of Trust Centralized or Decentralized? To address this question, we analyze reference wallet solutions, and mark them as centralized "PKI-based" or decentralized "DPKI-based".

RQ2. How are Credentials Shared in a Controlled Way? After understanding the trust model of the wallets, we analyze the selected solutions in terms of technologies used for controlled and trustworthy credential sharing. This analysis focuses on credential creation, management (including storage on mobile or web), sharing, and revocation.

3 Wallet Requirements and Technology Framework

Section 3.1 lists the general requirements necessary for realizing a digital identity wallet solution with a focus on the requirements of the recent European Digital Identity Wallet (EUDI Wallet) proposal in the context of electronic Identification and Authentication and trust Services 2 (eIDAS2) regulation[2] [20]. Section 3.2 then outlines the key technologies used to meet the wallet's requirements.

3.1 Requirements Elicitation

Functional Requirements. These requirements describe the functionalities that a digital wallet should implement. We conducted keyword research on digital wallets, particularly digital identity wallets, and identified that the main functional requirements for which they are developed are to store, manage, and share the identity or its related data. Manage (or controlling) the data includes removing and reviewing the data stored in the wallet and explicitly selecting what data to store/combine and share into/outside the wallet. Identity wallets also provide secure storage for cryptographic material associated with the data. The wallet may additionally have functionalities for recovery and back up [35,57,60]. The eIDAS2 regulation counts several functionalities as well for the EUDI Wallet in a recently published Architecture and Reference Framework (EUDI-ARF) [18]. The EUDI-ARF covers the aforementioned requirements along with other functionalities such as the creation of (qualified) electronic signatures/seals that demand the provision of interfaces to the services providing the functionality, requiring identification with an eIDAS high LoA.

Non-Functional Requirements. These requirements describe how a digital identity wallet satisfies a system's non-functional aspects [56]. Christopher Allen first in his paper [5] provided a comprehensive list of these requirements for a

[2] The eIDAS2 regulation aims to enhance the security and usability of electronic identification and trust services in the EU. The EUDI Wallet is expected to contribute to this objective by providing a standardized and interoperable digital identity solution.

user-centric wallet adhering to the Self-Sovereign Identity (SSI), referred to as SSI principles, including existence, control, access, and so on. The Sovrin working group provided later a rich list of these requirements in [65], which were categorized into three groups of "Agency", "Autonomy" and "Integrity". While there is some overlap between them, they are not identical. Both sets of principles emphasize the importance of control, access, transparency, and portability in SSI, but Sovrin's principles also prioritize equity, inclusion, and decentralization. See Table 1 for the definitions of each requirement. A user-centric digital identity wallet solution might fully or partially adhere to this list of requirements. The EUDI-ARF includes several aspects for the EUDI Wallet regarding these SSI principles as well and shows compliance with SSI, except for decentralization. This comes from the nature of trust in the context of eIDAS2. Although the eIDAS2 is technology-agnostic and supports both centralized and decentralized digital identities, acquiring qualified certifications for the services provided to the wallet from third-party providers or for the wallet itself is a limiting constraint [64].

3.2 Reference Technology Framework

The Trust over IP (ToIP) Foundation provides an open and collaborative platform for defining a complete architecture to establish digital trust between peers over the Internet or other networks. To achieve this, it combines available standards, protocols, and capabilities produced by standard development organizations and industry foundations such as the World Wide Web Consortium (W3C) and the Decentralized Identity Foundation (DIF) to fulfill the requirements of a digital trust architecture [23]. We based our evaluation of the technical foundation of the wallet solutions on the ToIP technology architecture or framework [3], which served as our inspiration. Additionally, we provide information in Table 1 on how and which technologies can be used to meet the non-functional requirements of digital identity wallets, along with a mapping of these requirements to the ToIP layers (find the layer numbers in the last column).

ToIP Overview. ToIP has four layers: trust, agent, service, and application. The first two layers focus on establishing machine-to-machine trust, enabling secure and private communication between machines. Public key cryptography plays a crucial role in these layers, allowing entities to verify each other using their identifiers. These identifiers can be centralized or decentralized, and the trust establishment can be done with or without a root of trust and through peer-to-peer protocols.

Layer 1 (trust) includes services designed to specifically support trust between peers in the upper layers, including trusted computing, secure storage of cryptographic materials, and trusted transport protocols such as HTTP(s), Bluetooth, NFC, and QR. The layer may also require support services that are outside the boundary of an application, such as trust registries, which offer functionality for discovering data about trustworthy entities (i.e., issuers and verifiers), applications, credentials (format, schema, and revocation mechanisms), identity assurance level, etc. These services can be implemented in various ways, using

Table 1. Non-functional requirements

	Requirement taken from [65]	Meaning	How it can be technically fulfilled	ToIP Mapping
Agency	**Representation**	Identity subjects must be able to be represented by any number of digital identities.	Subject identifiers, centralized Public Key Certificates (PKCs), or DIDs are used to achieve this principle. Identity subjects may also obtain multiple identifiers and build new identities when required. Additionally, they may prove their identity ownership in a one-way using zero-knowledge proofs, without including an identifier in the identity data (e.g., Attribute-based (or Anonymous) Credentials [63]).	1
	Delegation (of control)/Provability	Identity subjects must have the ability to control how their identity data is used, and exert this control by employing and/or delegating to agents and guardians of their choice, including individuals, organizations, devices, and software.	This is done by using asymmetric cryptography authentication protocols, such as DID authentication occurring over the DIDComm messaging protocol [26], to prove control over a proof private key linked to subject identifiers.	1, 3
	Equity and inclusion	Within a governance scope, identity subjects must not be subject to discrimination.	This is addressed by using an inclusive design approach to "all people regardless of their abilities, age, economic status, education, geographic location, language, etc. [76]," and by adapting user interfaces to satisfy the varied needs of different identity subjects [44].	4
	Usability, accessibility, and consistency	Agents and other IdM system components must be as usable and accessible as possible for identity subjects, providing a consistent user experience.	Because they are always accessible, mobile wallets are ideal for constant accessibility. Careful design choices must also be made to improve usability using relevant technologies [46].	4
Autonomy	**Participation (or consent)**	Identity subjects cannot be made to participate against their will. They must have the option to opt out of giving their identity data for services where it is not required.	To guarantee that the identity system has the data subject's consent and conforms with legal requirements like the EU GDPR, careful design decisions must be made [19]. For example, personal data must not be written in a public ledger, as anybody can read the data kept in a public ledger. Alternatively, the data registry can be implemented using a private permissioned ledger.	3, 4
	Decentralization	A subject's digital identity data must not be represented, controlled, or verified by an identity system that is dependent on a centralized system.	A decentralized trust paradigm is used to achieve this.	1
	Interoperability	Digital identity data for a subject must be represented, exchanged, secured, protected, and verified in an interoperable manner.	This occurs on several levels, including technical interoperability, which enables interoperability between various user-centric IdM systems using open standards and protocols, as well as between existing conventional centralized or federated IdM systems. Semantic interoperability, and organizational interoperability that aims for, say, compatibility between various blockchains and user-centric ecosystems [81].	2
	Portability	Identity subjects must be able to take their digital identity data anywhere (i.e., to the agents or systems of their choice).	Through the use of their private key, subjects are able to move their data between other devices and systems using distributed ledgers or blockchains [61].	2
Integrity	**Security**	Identity subjects must secure their identity data.	To protect data in transit and at rest, certain design considerations must be made, such as encrypted storage, signed transactions, and secure connections.	1-4
	Verifiability and authenticity	Identity subjects are required to provide verifiable proof of the authenticity of their digital identity information.	This is accomplished through the use of verifiable credentials and presentations.	1, 3
	Privacy and minimal disclosure	Identity subjects must protect the privacy of their digital identity data and share the minimum digital identity data required for any interaction.	User-centric IdM systems are privacy-oriented and provide the identity subject full control. The privacy features they give, however, differ depending on the credential types and proof they provide.	1-4
	Transparency	Identity subjects and all other stakeholders must have easy access to and verify information necessary to understand the incentives, rules, policies, and algorithms under which agents and other components of the ecosystems operate.	This is achieved by developing wallet solutions based on open protocols and open standards.	1, 3

distributed ledgers that host the registries or identity federations that treat only the members of the federation as trustworthy.

Layer 2 (agent) includes protocols that serve as a bridge between higher-level application-facing protocols and lower-level transport protocols. ToIP requires that these protocols define how to construct and format messages that can be verified to have certain properties, such as authenticity and integrity [2]. Examples of protocols that meet these requirements are OpenID for Verifiable Credentials (OpenID4VC) [31] and DIDComm messaging [26]. OpenID4VC protocols, for instance, satisfy these properties and define APIs for the issuance and presentation of credentials in the upper layer.

Layer 3 (service) is responsible for defining and executing the trust services required for establishing trust between different entities in the ToIP ecosystem. It includes functions such as verifying identities, establishing credentials, and authorizing access to resources. In our analysis, we focus on services related to the credentials lifecycle [75], including the issuance, request and presentation of credentials, as well as verification and revocation in this layer. We first explore the credentials-related type/data model, which mainly consists of metadata, claims (or identity attributes), and proofs under the relevant trust model. The relevant technologies often come in the form of libraries that implement different types of credentials, such as W3C VC [75], ABC [63], and mDOC [41].

Next, we explore the credential issuance or presentation protocols, which may run on top of Layer 2 protocols or use another service within this layer. For example, in the case of OpenID4VC, it occurs through specific protocols such as OpenID for Verifiable Credential Issuance (OpenID4VCI) [77] and OpenID for Verifiable Presentations (OpenID4VP) [78], or through DIDComm messaging [26], supporting Wallet and Credential Interaction (WACI) protocols such as the Issue Credential Protocol [29] and Present Proof Protocol [28], W3C Credentials Community Group VC APIs [73], Credential Handler API (CHAPI) [4], ISO mDOC Request and Response Protocol [42], and so on. Verification of credentials enables a wallet to verify and prove the authenticity and integrity of data in transmission by implementing asymmetric cryptography algorithms such as Digital Signatures (DSs) or Zero-Knowledge Proofs (ZKPs). Available technologies can be provided as a library, for example, Hyperledger Ursa [30] or Idemix [36], or as a service, such as an eIDAS trust service like an electronic signature [17].

Revocation allows for checking the validity of a credential before trusting it for authentication. Various revocation mechanisms are available, including Accumulator Value (AV) [14], Credential Status List [27], Validity attestations, and others.

We are also interested in identity-binding mechanisms. These mechanisms, such as token binding [62], are used to establish a connection between a credential subject and either the application or the credential itself. The authentication of binding parameters linked to the credentials allows the issuer (or verifier) to verify the credential subject and issue (or accept) the credentials. The authentication performed in this layer is separate from the public key authentication mentioned earlier in Layer 2. Its purpose is to verify that the current subject is indeed the intended recipient of the credentials. Biometric traits are one of the

methods used for this purpose, as they offer individuals an immutable characteristic, thereby providing a high LoA authentication.

And finally, Layer 4 (application) facilitates the interaction between individuals (or credential subjects) and applications, allowing them to engage in trusted interactions that serve specific business, legal, or social purposes.

4 Wallet Analysis

First Wallet Analysis. We conducted a Google search using the terms "Digital identity wallet" or "SSI wallet" and found a significant number of wallets. As discussed in [45], these wallets were developed by various entities, including big tech companies such as Apple, public sectors such as the European Union, and independent wallet providers and startups like Evernym. These wallets vary in terms of their use cases, target audience, security, privacy, interoperability, and many other features as mentioned in Sect. 3.1, as well as the underlying technology stack. To account for these differences, we collected a set of 32 wallet solutions, called as "Reference Wallets" to form the basis of our study, and anlayzed them first based on six questions that arose during our search:

1. What kind of solution is it? Is it a WEB, or a mobile (MOB) solution?
2. What is the solution use case (UC)? Is it for personal (PERS) or business (BIZ) use cases? Personal ones are designed for individuals to manage their digital identity in a secure and convenient way. While the business ones on the other hand tend to have more specific features and requirements that are tailored to the needs of businesses and organizations, such as to store and manage employee data, such as HR records and payroll information.
3. Who is responsible for the wallet development? Is it a public (PUB) or private (PVT) solution? A public (or national) wallet is funded and developed by public sectors (or in partnership with a few private companies or organizations). A private wallet, on the other hand, is offered by one or more private sector (profit/non-profit) organizations, that are primarily in charge of wallet development.
4. Is the source code of the solution available online (OSS)?
5. What is the maturity level (MAT) of the solution? Is it a field product (FLD), a PILOT (tried with a small group of real users), a prototype PROTO (a working model of the solution with limited functionality), or a DEMO (a presentation of the solution)?
6. What is the employed trust model (TRU MDL) and if it uses a blockchain, which kind of blockchain is it?

Table 2 reports the reference wallets and their analysis.

Second Wallet Analysis. For the second analysis, we narrowed down the list of reference wallets to 12 (Connect.Me, KayTrust, Talao, Apple, IRMA, IDgov.pt, Open, mObywatel, IDEMIA, DIZME, Microsoft Authenticator, and Verse)-roughly 38% of them, based on their trust model, ensuring that the

Table 2. Reference wallets

No	Name	Type		UC		Developer	OSS	MAT	TRU MDL	Blockchain
		Web	MOB	PERS	BIZ					
1	Trinsic	Y[1]	Y	Y	Y	PVT	Y	FLD	DPKI	Sovrin, Indico (Indy networks)
2	Connect.Me	-	Y	Y	-		Y	FLD	DPKI	Sovrin
3	Lissi	-	Y	Y	-		Y	FLD	DPKI	Sovrin, IDUnion
4	esatus	-	Y	Y	Y		Y	FLD	DPKI	Sovrin, IDUnion
5	Jolocom SmartWallet	-	Y	Y	Y		Y	FLD	DPKI	Ethereum
6	Shocard	-	Y	-	Y		N	FLD	DPKI	Bitcoin
7	KayTrust	-	Y	-	Y		Y	FLD	DPKI	Alastria
8	SelfKey	-	Y	Y	-		Y	FLD	DPKI	Ethereum
9	Data	-	Y	Y			Y	FLD	DPKI	EBSI networks
10	Gataca	-	Y	Y	-		-	FLD	DPKI	Ethereum and EBSI networks
11	Talao	-	Y	Y	Y		Y	FLD	DPKI	Tezos, EBSI networks
12	AceID	-	Y	Y	Y		Y	FLD[2]	DPKI	AceBlock
13	Osma	-	Y	Y	Y		Y	FLD[3]	DPKI	Bitcoin, ION
14	DataKeeper	-	Y	Y	Y		N	FLD	DPKI	Ethereum Quorum
15	Microsoft Entra Verified ID	-	Y	Y	Y		Y	FLD	DPKI	Bitcoin
16	My Optus (Mastercard ID)	-	Y	Y	Y		-	PILOT[4]	DPKI	ID Network
17	Bloom	-	Y	Y	Y		Y	FLD	DPKI	Ethereum
18	DID	-	Y	Y	Y		Y	FLD	DPKI	AceBlock
19	ID Pocket	-	Y	Y	Y		N	FLD	DPKI	Ethereum
20	IDnow Online Ident	-	Y	-	Y		N	FLD	DPKI	IDUnion
21	IRMA	-	Y	Y	Y		Y	FLD	PKI	-
22	VIDwallet	-	Y	Y	Y		N	FLD	DPKI	Alastria, EBSI networks
23	Walt.id Wallet	Y	Y	Y	Y		Y	DEMO[5]	DPKI	Gaia-X/ Ocean, EBSI networks
24	DIZME	-	Y	Y	Y		Y	FLD	DPKI	Sovrin
25	Apple wallet	-	Y	Y	-		N	FLD[6]	PKI	-
26	ID Wallet	-	Y	Y	Y	PUB	Y	PILOT[7]	DPKI	Hyperledger Indy
27	Alastria ID	-	Y	Y	Y		Y	DEMO[3]	DPKI	Alastria networks
28	Open wallet	-	Y	Y	-		Y	PROTO[3]	DPKI	Trustchain
29	IDEMIA (Mobile ID)	-	Y	Y	-		N	FLD	PKI	-
30	Veres[8]	Y	-	Y	Y		N	FLD	DPKI	EBSI networks
31	mObywatel	-	Y	Y	-		N	FLD	PKI	-
32	IDgov.pt	-	Y	Y	Y		N	FLD	PKI	-

[1] Trinisc Wallet SDK makes it possible to embed wallet functionalities into a web application.
[2] It has recently been removed from Google Play.
[3] Only the source code is available on GitHub.
[4] Currently launched in Australia (using My Optus wallet) and Brazil, see https://idwall.co/en/ for Brazil.
[5] Walt.id Wallet Kit enables re-branding, extending their web wallet[2], or building a new web or mobile wallet application.
[6] Currently, only the US states of Arizona, Colorado, and Maryland support IDs in the wallet.
[7] The ID wallet was temporarily removed from the App Store in September 2021 to improve it with respect to several aspects, such as the technical implementation and security of the wallet.
[8] Not accessible anymore.

included solutions either use a PKI or a DPKI model. As can be seen in Table 2, only 5 out of the 32 wallet solutions used PKI. We included all PKI-based solutions in the second analysis. From the DPKI-based solutions, we selected two from public solutions and the rest from private ones. This provides us with an approximate 40–60 percent rate for the trust models in developer types. If a DPKI-based solution with a blockchain in the system is present, we attempted to include various blockchains in the analysis. Find the selected wallet solutions highlighted as green in Table 2.

4.1 Wallets Description

DPKI-Based Wallet solutions:

- Connect.Me is the first SSI mobile wallet built by Evernym on the Sovrin network [66], which uses a public permissioned blockchain of Hyperledger Indy and will work with cheqd network in the future [12]. It can be adjusted to a variety of use cases, enabling access to both public and private services as well as the sharing of identity-related data in the form of Anonymous Credentials with the privacy features it offers.
- KayTrust is a mobile wallet developed by Everis and allows users to handle digital credentials for a variety of purposes, including diplomas, vaccination credentials, labor certificates, land registration documents, and any other digital accrediting document. The wallet improves the accreditation and certification process, combats document fraud, and ensures security, transparency, and digital truthfulness by allowing a credential to verify its authenticity, origin, and validity. At the same time, it seeks to improve the user experience, which will carry out real-time verification on the registration in the LACChain Ethereum network [6].
- Talao [10] is the first French mobile wallet adapted to the European market, developed by the Talao European startup. It is the first wallet to establish identity data in the format of the Tezos network [8]. Talao follows the recommendations of the European Self-Sovereign Identity Framework (ESSIF) to comply with the EU Digital Identity toolbox [18]. It enables users to collect various documents such as attestations, certificates, membership cards, permits, and identity passes, which serve as proof of identity and rights for accessing online services. User data is stored on the user's smartphone and exchanged in accordance with the GDPR [19]. This is achieved through the use of the DID web method, which does not require writing DIDs on a blockchain and hosts them and the related documents on a web domain [34]. Additionally, user consent is explicitly requested for certain actions.
- Open is a prototype wallet that was developed as part of an effort by the Dutch government to build an SSI system over the IPv8 peer-to-peer networking protocol [69], leveraging Trustchain [70]. It provides a REST API to retrieve and share identity data online or offline using Bluetooth, NFC, and QR technology. IPv8 also shares the data in a zero-knowledge way using three different protocols, namely Peng-Bao range proofs [59], Camenisch-Lysyanskaya (CL)

signatures, and an identity-based proof of plaintext [32] using Boneh's cryptosystem [11]. Additionally, the wallet stores identity data in an encrypted form, and its decryption necessitates passport facial recognition [68].

- DIZME (this is me) is a project by the Trust over IP Foundation aiming to fill the gap between SSI and eIDAS. This project details the governance and technology utilized to achieve certain LoA levels. DIZME achieves LoA levels up to substantial. DIZME combines the benefits of SSI, with the legal value and trust of a Qualified Trust Service Provider (QTSP), leveraging InfoCert [67], Italy's first certification authority, which issues and manages qualified digital signature certificates. InfoCert itself acts as a founding steward of the Sovrin trust framework. In brief, the Sovrin ledger is operated by Stewards, trusted organizations within the ecosystem who have agreed to abide by the requirements of the Sovrin trust framework and are responsible for operating the nodes that maintain the Sovrin distributed ledger. Interoperability between SSI and the trust services defined by the eIDAS regulation is made possible by the dual role of InfoCert, which acts not only as founding steward of the Sovrin network (as well as the DIZME governance authority) but also as a QTSP under the eIDAS regulation. DIZME supports both full-disclosure and ZKP formats. Besides depending on the credential LoA presented by the user, the verifier can issue a signature request (advanced or qualified) to obtain a digital signature (one-shot and eIDAS compliant) with which the user can confirm a specific transaction, fully guaranteeing legal compliance and value.
- Microsoft Authenticator was upgraded and transformed into an SSI mobile wallet as of August 2022. It enables its users to manage VCs through the Microsoft Entra Verified ID service [58]. A service that supports available SSI standards such as W3C VC [75], DIDs [74], and OpenID4VC [80]. The wallet obtains and presents the credentials using OpenID4VC protocols [80] in JSON Web Token (JWT) format with an Edwards-curve or elliptic curve digital signature. The signature is verified by resolving entities' DID documents recorded on Bitcoin (leveraging the ION Identity Overlay Network [25]) or a web domain. The ION is a Layer 2 open, permissionless network that relies on the Sidetree protocol to batch transactions together before recording them on the Bitcoin ledger. This resolves the problem of increasing transaction time on the ledger [21]. By batching, the network nodes can write a large number of DID documents to the Bitcoin ledger in a single transaction.
- Verse is an SSI web wallet that has emerged as a result of the EBSI4Austria project [9]. This project, funded by the Connecting Europe Facility (CEF), aims to set up, operate, and maintain Austria's EBSI node and pilot the diploma use case in the country. EBSI4Austria consists of three partners, namely two universities, the Graz University of Technology (TU Graz) and the Vienna University of Economics (WU Vienna), along with Danube Tech, a Vienna-based company specializing in SSI and distributed systems, which is actively involved in related standardization bodies. The Austrian EBSI node is set up and operated at the Department of the eGovernment Innovation Center (EGIZ), which is a part of the TU Graz.

PKI-Based Wallet Solutions

- IRMA is an SSI mobile wallet produced and maintained by the Privacy by Design Foundation (pbdf) [24]. It is used for identification and digital signing purposes. The IRMA SSI solution implements the Idemix cryptographic protocol [36] and issues Attribute-Based Credentials (ABCs). Instead of relying on a blockchain, the system trusts an IRMA server that hosts issuer public key information, which is operated by a verifier in the system. The wallet holder shares the ABC credentials in a zero-knowledge way by implementing the CL signature [15], see [40] for details.
- Apple has launched a collaboration with various US states to enable their residents to link their driver's license or state ID to the wallet on their iPhone and Apple Watch [39]. Arizona and Georgia were the first states to offer this technology to their citizens, followed by Connecticut, Iowa, Kentucky, Maryland, Oklahoma, and Utah. The Apple wallet adopts the International Standards Organization (ISO) 18013-5 standard[3] for Mobile Driving Licence (mDL) [42], which supports centralized PKI and guarantees privacy features such as selective disclosure.
- IDgov.pt is a mobile wallet launched by the Administrative Modernization Agency in Portugal. It stores and periodically updates (every 24 h) digital documents issued by the Portuguese State, such as citizen cards, driving licenses, and ADSE cards. This update occurs after successful user authentication with a Mobile Digital Key (CMD) through a national interoperability platform that makes the data available. Users can share their documents offline by generating a QR code that can be scanned by a similar wallet application and validated through an authentic source of data or by using a validity code via the government portal[4]. The wallet will adopt to the ISO 18013-5 standard and support the Bluetooth protocol in its future versions. Users can also generate a certified PDF that includes a Qualified Electronic Signature (QES) and share it via WhatsApp, email, etc.
- mObywatel is a mobile wallet launched by the Polish Ministry of Digital Affairs in 2018. It allows Polish citizens with a national ID card to access government services such as logging into the e-Tax Office, settling tax returns, storing confirmation of COVID-19 vaccination, verifying obtained driver's qualifications, checking driving license penalty points, and more (find a list of all the services in [48]). Different applications are used in conjunction with mObywatel to enable citizens to obtain their mobile documents. For example, to obtain mobile student ID cards (mLegitymacja [72]), university students need to use the Mobile USOS[5] which is the official university application for requesting

[3] Although ISO 18013-5 focuses on the mDL, it can be used to provide a variety of credentials.

[4] https://www.autenticacao.gov.pt.

[5] University Study-Oriented System (USOS) is a system that authorizes a student through the central authorization service (CAS) of the university, providing access to the student's educational information in the USOS database. Find more information at http://usos.edu.pl.

mobile student cards. The Mobile USOS forwards the request to the Ministry and displays a QR code and PIN to retrieve the information about the issued cards. The QR code and PIN are then copied and pasted into the mObywatel application to access the services. The credentials can be shared online via the login portal or using the mWeryfikatora application offline.

– Mobile ID is a user-centric mobile wallet by IDEMIA that allows American citizens to benefit from a government-issued trusted digital ID or mDL, enabling different services such as opening a bank account and filing taxes. Mobile ID adopts ISO 18013-5 [55] for mDL/ID and NIST SP 800-63-3 for IdM [55]. It enables identity verification online, offline, remote, and in-person [38]. NIST SP 800-63-3 specifies enrollment and identity proofing, as critical aspects for credential issuance, leveraging the Mobile ID wallet. This allows the wallet to derive and store identity data locally either from a governmental database or an eID document. The Mobile ID wallet integrates IDEMIA biometric check solutions (Capture SDK/Web Capture) that enable face verification, including liveness checks [37].
  Mobile ID wallets have recently been developed in partnership with US state partners such as the Arizona Department of Transportation, Delaware Division of Motor Vehicles (DMV), Oklahoma, and Mississippi Department of Public Safety (DPS). Additionally, Mobile ID has also been developed for the state of Iowa and Colombia[6]. IDEMIA has recently collaborated with Microsoft Azure Active Directory to deliver a secure digital verified credentials solution, which is currently in public preview [1].

4.2 Answer to Research Questions

We provide a technical synopsis of the wallet solutions for Layer 1 (trust), remote support for that, and Layer 3 (credential) of ToIP found in Table 3, as our response to the research questions.

Answer to RQ1: How is Trust Established? Is the Root of Trust Centralized or Decentralized? There are several trust services available with hardware and software support at Layer 1. To make the analysis manageable, we have only considered the trust support for implementing the root of trust, as well as the trust registry service. Accordingly, Table 3 shows how trust is established for both user-centric PKI solutions using CAs and DPKI-based solutions using a blockchain in the system. The technological basis of user-centric PKI-based solutions is largely based on the ISO 18013-5 (mDL) standard, or it is being considered for future development. On the other hand, the DIDs and VCs standards have served as the basis for DPKI-based solutions, known as user-centric SSI, up until this point. Our study confirmed that utilizing SSI has primarily been associated with using a blockchain. However, SSI has a reputation for being blockchain-agnostic. DIDs can be used across different networks and

[6] No applications were found for Colombia.

Table 3. Technical specifications for the study wallets

No				1	2	3	4	5	6	7	8	9	10	11	12
ToIP Layer			Tech Specs. / Wallet Name	Connect.Me	KayTrust	Talao	IRMA	Open	DIZME	Verse	Microsoft	Apple	IDgov.pt	mObywatel	IDEMIA
Credential	Credential Type		VC [75]		✓	✓				✓	✓				
			ABC [63]	✓			✓	✓	✓						
			mDOC [42]									✓			✓
			PDF										✓		
			QR code										✓	✓	
	Encoding Scheme		JSON [52]		✓		✓	✓		✓	✓		✓	✓	
			JSON-LD [51]	✓	✓	✓			✓	✓					
			CBOR [16]									✓			✓
	Exchange Protocol		DIDComm WACI	✓					✓					- No specific protocol	
			OpenID4VCI/ OpenID4VP [77,78]		✓	✓					✓				
			CHAPI [4]							✓					
			RestAPI				IRMA protocol	✓							
			mDOC Request/ Response [42])									✓[1]	-[2]		✓[1]
	Verification (or proof)	ZKP	ZKP-BBS+ [53]	✓		✓			✓						
			ZKP-CL [50]				✓	✓							
			ZKP- range & identity-based proof [32,59]					✓							
		DS	VC-JWT [75]		✓					✓	✓				
			VC-LD Signature [75]		✓					✓					
			PoP PKI/MSO [42]									✓		✓	✓
	Revocation		AV [14]	✓			✓		✓						
			Credential status List [27]			✓		✓[3]			✓				
			Revocation status		✓			✓		✓					
			Out of scope									✓	-	-	✓
	Authentication/ Binding		Biometric	FR[4]	UNLKG	UNLKG (OPT)	+ If rqd LivDet	LFR	In three LoAs[5]	-	FR	FR & FPR	-	FR & FPR UNLKG	LFR (or 6-digit PIN)
			PIN			UNLKG	✓						CMD		
			Password						✓				-	✓	
			Email		ACTIV					✓			-		
			Phone								✓				✓
Trust	Trust Registry		Trusted issuer	-	✓	-	✓[6]	✓[7]	-	✓	-	-	-	-	-
			Trusted verifier				-			POSS					
			Trusted application		- Holder			- Holder		✓					
			Credential schema	✓	✓	-	✓	Schema URI (OPT)	✓	✓	✓				
			Revocation	✓	-	✓	-[8]	-	✓	✓	-[9]				
	Root of Trust	Blockchain -based	Sovrin	✓					✓						
			EBSI (Besu, Fabric)							✓					
			Alastria (GoQuorum)		✓										
			Tezos (and Ethereum)			✓[10]									
			Trustchain					✓							
			ION (Bitcoin)								✓				
		Conventional/X.509 PKI					✓					✓	✓	✓	✓

FR: Facial Recognition, LFR: Live Facial Recognition, FPR: Fingerprint Recognition, UNLKG: Unlocking, ACTIV: Activation, OPT: Optional, POSS: Possible

[1] The credentials in Apple are derived from ID cards, while IDEMIA's credentials can be derived from ID cards or governmental datasets. Apple and IDEMIA applications only use the ISO-specific mDOC Request and Response protocol to present credentials to verifiers.

[2] Credentials can be shared over screen-camera communications by generating a QR code, or through a validity code via the government portal. Future versions are anticipated to adopt ISO 18013-5 and use the mDOC Request and Respond protocol over Bluetooth.

[3] Updated attestation status, presented by an attester (or issuer).

[4] In integration with iProov and Onfido.

[5] Authentication happens in three levels: 1) video selfie (and email or phone proof), 2) live photo (and ID card) and QES or 3) liveness assurance (remote or in person).

[6] Only the credential schema appears for trusted issuers in the GitHub repository of pbdf.

[7] All the entities must be registered in an AttestationCommunity, which provides a distributed dataset of their information.

[8] There is no specific revocation registry present. Revocation only happens by the holder and use of the IRMA keyshare server.

[9] By the use of the issuer Identity Hub API endpoint.

[10] Talao may also use a web domain.

platforms, offering better interoperability, enhanced security (no single point of failure), and no third-party control.

Answer to RQ2: How are Credentials Shared in a Controlled Way? This question explores how credentials are selectively shared or disclosed to meet specific requirements or predicates. Wallets employ various data disclosure protocols and cryptographic proofs to enable a controlled sharing of credentials. Table 3 provides an overview of the credential types, encoding schemes, exchange protocols, relevant proofs used for verification, and the revocation mechanisms applied to these credentials.

Table 3 shows credential types/data models mainly as W3C VCs and ABCs, or ISO 18013-5 mDOCs. ABCs are the most privacy-preserving credentials. They enable selectively disclosing data while preserving the privacy of the user using ZKPs[7] [33], and by avoiding any correlation between the credentials and their subjects. On the other hand, VCs are less privacy-preserving compared to ABCs and only a few proposal protocols exist to enable selective disclosure on them [22,79]. Find more detailed information on different types of VCs and their variations ABCs in [82]. ISO similarly offers an mDOC credential type with Concise Binary Object Representation (CBOR) or JavaScript Object Notation (JSON) encoding for sharing credentials offline or online, respectively. The mDOC credential includes a signed Mobile Security Object (MSO) as proof, supporting selective disclosure of the mDL credential. The MSO contains digests of each mDL attribute. The related public key of the signer/issuer, linked to its identity using X.509 PKC, must be maintained by a CA, which is a Verified Issuer Certificate Authority List (VICAL) provider in the mDL context [42]. Besides, the mDL credentials are linked to the holders' identifiers, typically their public keys associated with their wallets. This linkability enables the traceability of the mDLs back to their holders, which can potentially compromise their privacy.

Biometric authentication is a popular method employed by wallets for either binding credentials to individuals or for wallet authentication. For instance, ISO supports the biometric binding of mDL credentials by incorporating a user portrait within the credentials.

Revocation of a DPKI-based wallet credential is based on the data model and proof format, which, for instance, may use an accumulator value for those supporting a ZKP. On the other hand, ISO only adds a validity date to the credentials; revocation is not supported and may be implemented in practice through additional means outside the ISO standard.

User-centric PKI-based wallet agents predominantly exchange mDOC requests offline. However, ISO also defines OpenID or WebAPI protocols for online communications. On the other hand, the most significant credential exchange protocols in DPKI-based cases are DIDComm Messaging, which builds on the decentralized architecture of DIDs, and OpenID family protocols in the

[7] ZKPs enable a prover to persuade a verifier of the validity of a mathematical statement without revealing any further information.

context of VCs, namely OID4VCI and OID4VPs [77,78]. These OpenID protocols are credential-agnostic and also support the mDOC data model, allowing for alignment between the VC and the ISO-compliant mDL data models. This alignment presents interesting possibilities, particularly in the context of eIDAS2. Both technologies are included in the EUDI-ARF version 1.0.0, opening up new avenues for research in digital identity wallets. Notably, organizations like Spruce are actively working to align these standards, as mentioned in their work [13]. Furthermore, several solutions supporting technologies like OpenID Connect ensure interoperability with existing IdM systems. This concept has been explored in a study by Kuperberg et al. [49], which presents patterns and approaches for seamless integration between traditional IdM systems and associated applications with user-centric solutions.

5 Conclusion

In this paper, we have reviewed the technological trends of user-centric digital identity wallets, considering the trust models they employ and the control they provide. By formulating two research questions and adopting the ToIP layered technology framework, we have explored the most widely used technologies that address these questions. Each layer of the framework contributes in distinct ways to these dimensions, from the creation of tamper-proof digital credentials to enabling secure and selective credential exchange within the system. This investigation aims to assist designers in developing solutions that offer better functionalities including enhanced security, privacy, and interoperability to users, thereby promoting adoption. Nevertheless, identity wallets are still relatively new, and their adoption rates remain low. Future research in this area could focus on understanding the barriers to adopting these solutions, such as usability issues or lack of trust, and developing strategies to overcome these barriers.

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